CREDIT RISK RESEARCH CAPSTONE

Predictive Modelling, Calibration, Explainability and Fairness Assessment

Dataset: UCI Default of Credit Card Clients dataset (historical Taiwan credit-card data).

Objective: Evaluate machine-learning approaches for identifying customers who may default on future credit-card payments, considering predictive performance, probability quality, threshold selection and subgroup evaluation.

Report generated: 11 October 2026

This report summarizes the analyses and saved results available in the project at generation time. It is an educational research exercise, not a validated lending system or a recommendation for individual credit decisions.

1. Executive Summary

This project investigates credit-card default prediction using logistic regression, random forest and gradient boosting. The workflow includes data validation, exploratory analysis, model comparison, calibration assessment, SHAP-based explanation, threshold analysis and subgroup evaluation.

In the saved final evaluation, gradient boosting achieved a ROC-AUC of approximately 0.7849 and average precision of approximately 0.5517. Its Brier score was approximately 0.1342. These results describe this dataset and evaluation procedure; they do not establish performance on future customers or other populations.

The operating threshold affects the trade-off between detecting defaults and producing false-positive predictions. Subgroup estimates also differ and have varying uncertainty. These findings support further validation rather than a conclusion that the model is fair or ready for operational use.

2. Dataset and Data Validation

The working dataset contains 30,000 rows and 24 columns, including 23 input features and the target variable *default_payment_next_month*. The observed default rate is 22.12%. The project validation reported no missing values and 35 exact duplicate rows.

The working CSV does not include the original customer ID. Consequently, identical rows cannot automatically be assumed to represent the same individual, and duplicate rows were retained. The historical Taiwan context should be considered when interpreting these results.

Dataset validation summary

Validation check	Result
Dataset	Default of Credit Card Clients
Source	UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/350/default
Raw data file	data\raw\credit_card_default.csv
Processed data file	data\processed\credit_card_default.csv
Rows	30000
Columns	24
Missing values	0
Exact duplicate rows	35
Duplicate rows removed	0
Observed default rate (%)	22.12%
All validation checks passed	True

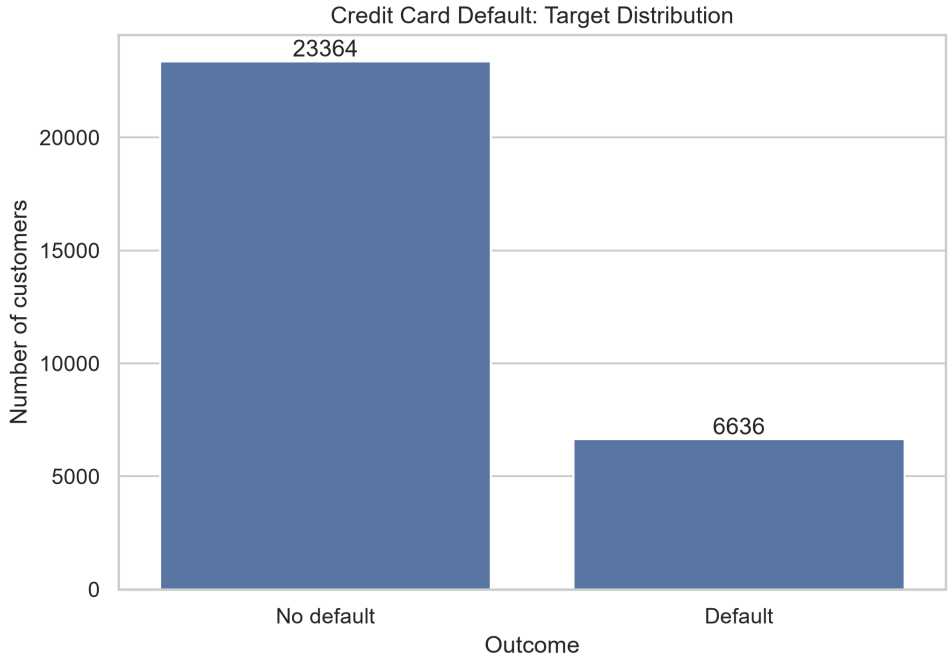


Figure 1. Distribution of the target classes.

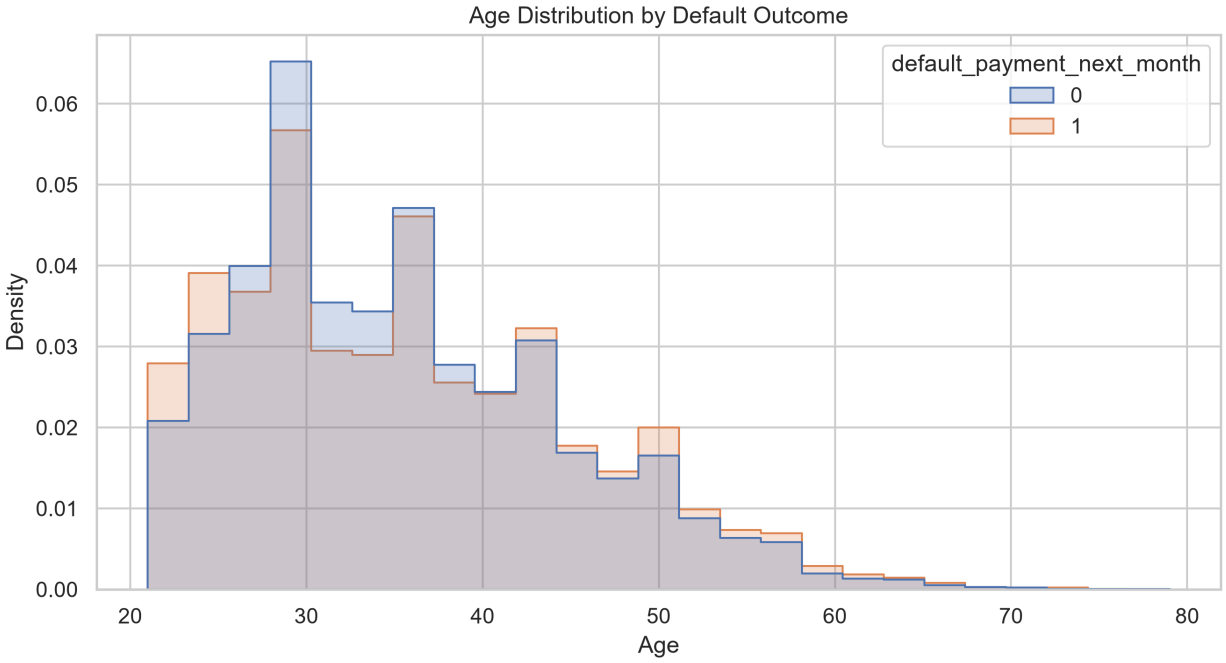


Figure 2. Age distribution by observed default outcome.

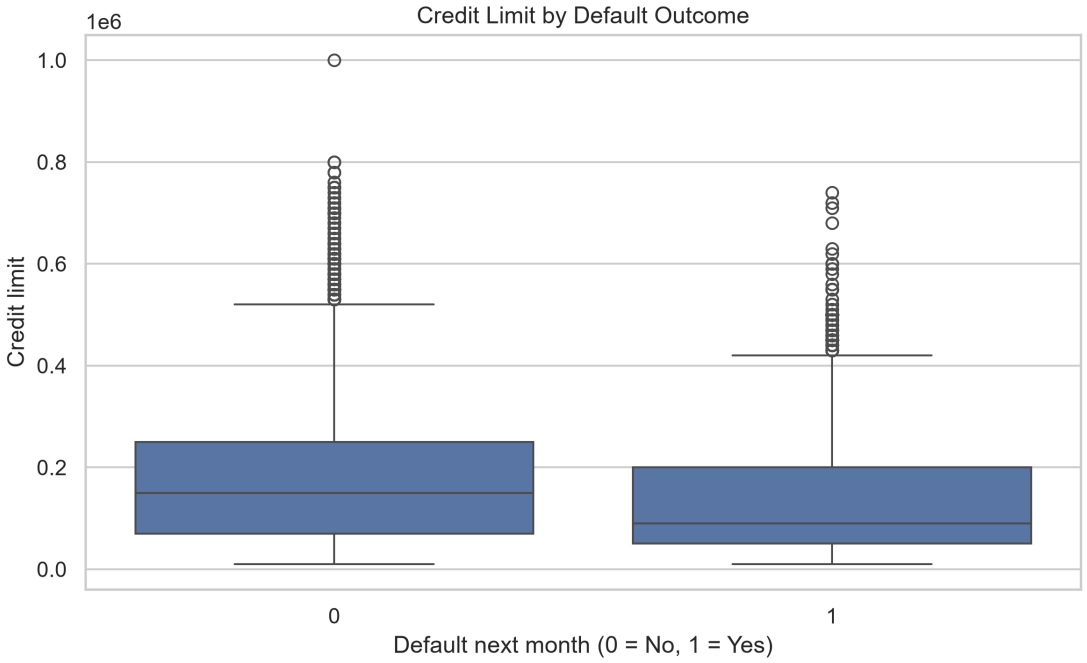


Figure 3. Credit-limit distribution by observed default outcome.

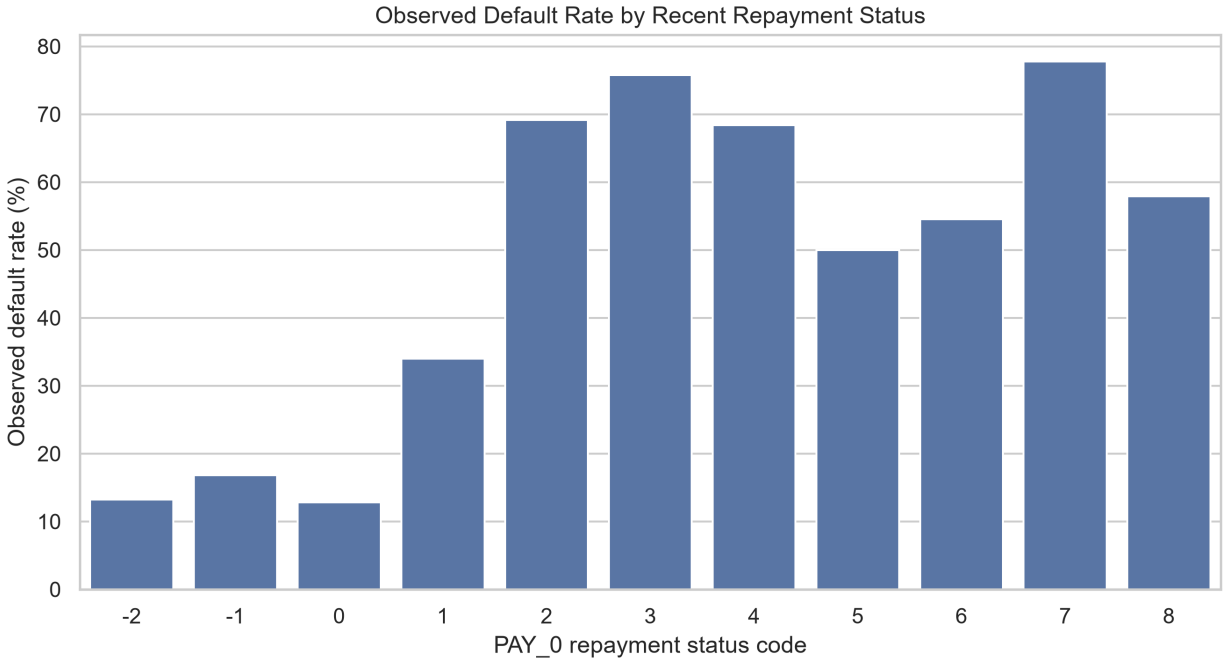


Figure 4. Observed default rate by PAY_0 repayment-status category.

3. Methodology

Data preparation: The project checks column names, missing values, duplicate rows, target values and basic data distributions.

Model development: Logistic regression, random forest and gradient boosting are compared. Categorical variables are encoded and numerical variables are scaled in the logistic-regression pipeline.

Evaluation design: The final evaluation script uses a stratified 60% training, 20% validation and 20% test split, with random seed 2026. Model selection is based on validation ROC-AUC. The selected model is then retrained on training plus

validation data before final test evaluation.

Metrics: ROC-AUC, average precision, accuracy, precision, recall, F1, Brier score and log loss evaluate different aspects of model performance. No single metric fully describes the practical suitability of a credit-risk model.

Important qualification: Earlier exploratory modelling used the same dataset before the final split was introduced. The final test set is therefore not an entirely untouched external evaluation. Independent external validation would provide stronger evidence of generalization.

4. Final Model Evaluation

The following table summarizes the saved final test results. Higher ROC-AUC and average precision indicate better ranking performance, while lower Brier score and log loss indicate better probabilistic predictions.

Table 1. Final test-set model comparison

Model	ROC-AUC	Avg. precision	Brier score	Log loss
Logistic Regression	0.7376	0.5142	0.1432	0.4603
Random Forest	0.7781	0.5465	0.1393	0.4401
Gradient Boosting	0.7849	0.5517	0.1342	0.4287

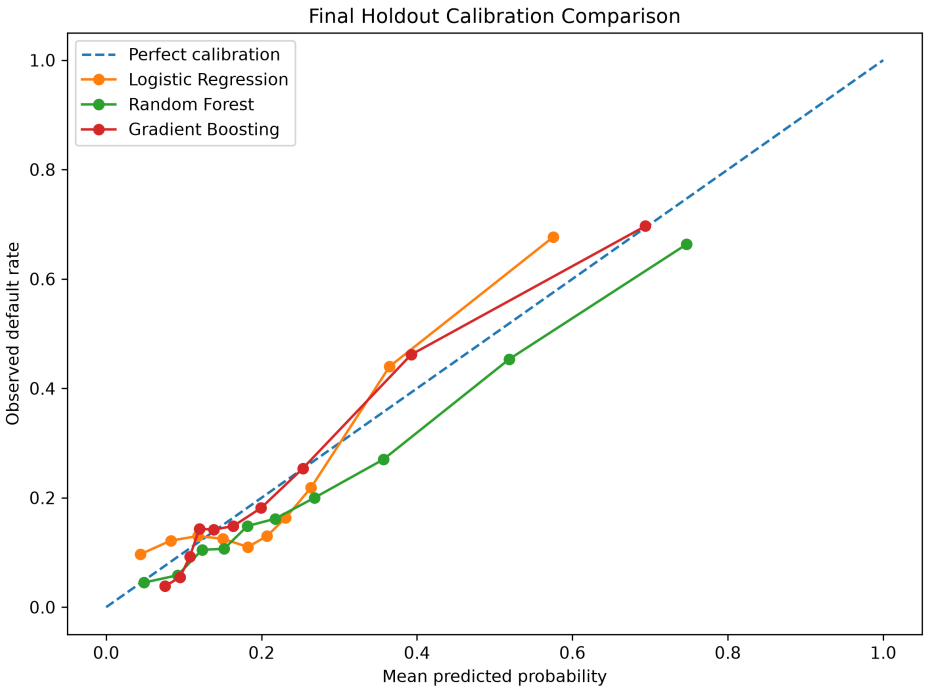


Figure 5. Calibration of predicted probabilities on the final test set.

5. Model Explainability

SHAP was used to examine which features influenced the gradient-boosting model's predictions. The saved analysis ranks PAY_0 among the most influential features, followed by LIMIT_BAL, PAY_3 and several payment-amount variables.

Feature importance describes model behaviour and associations in the analysed data. It does not establish causality, prove that a feature is appropriate for lending, or demonstrate that the model is free from unfair effects.

Leading SHAP feature-importance results

Feature	Mean absolute SHAP
numeric__PAY_0	0.4715
numeric__LIMIT_BAL	0.1459
numeric__PAY_3	0.0818
numeric__PAY_AMT3	0.0809
numeric__PAY_AMT2	0.0750
numeric__PAY_2	0.0706
numeric__PAY_AMT1	0.0692
numeric__BILL_AMT1	0.0489
numeric__PAY_6	0.0488
numeric__PAY_AMT6	0.0484
numeric__PAY_4	0.0467
numeric__PAY_5	0.0463
categorical__MARRIAGE_2	0.0254
numeric__PAY_AMT5	0.0224
numeric__PAY_AMT4	0.0207

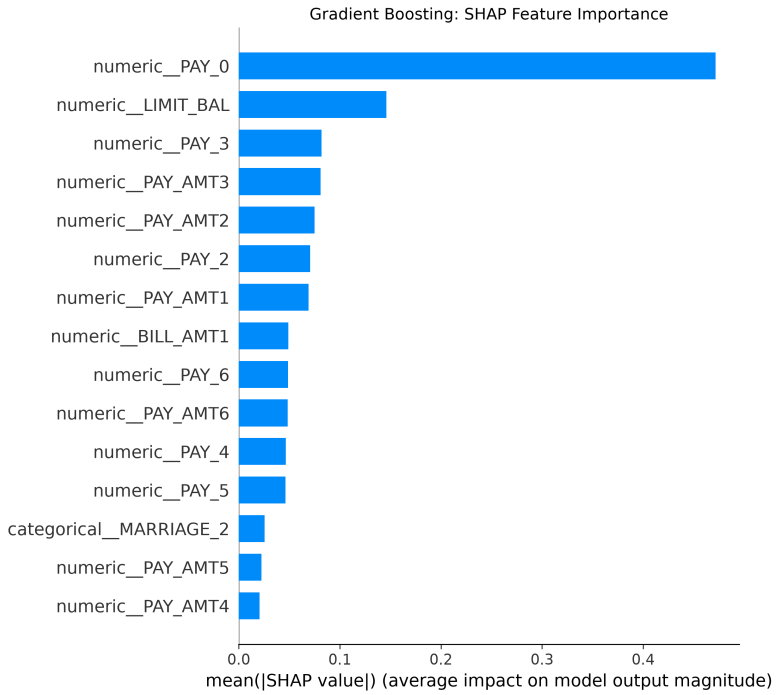


Figure 6. Mean absolute SHAP feature importance.

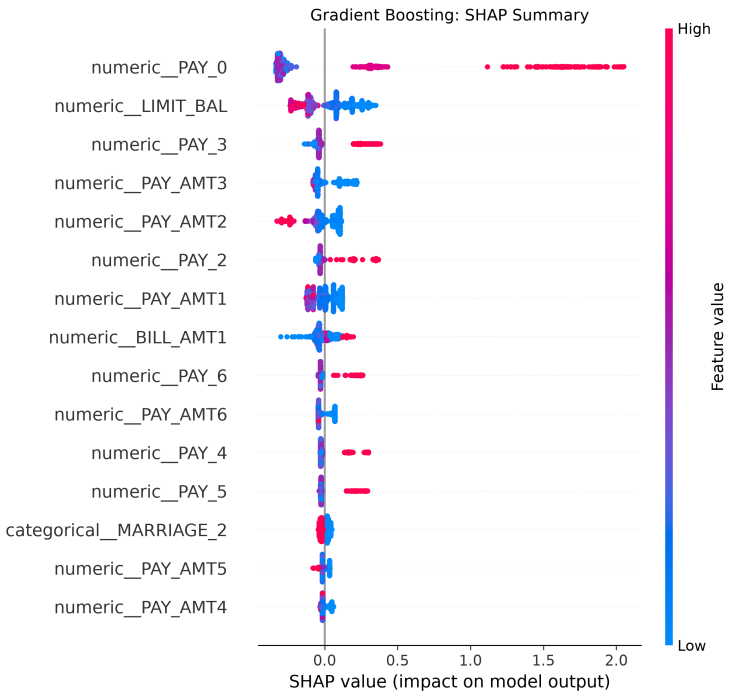


Figure 7. SHAP summary of feature contributions.

6. Decision Threshold Analysis

A probability threshold determines when a prediction is labelled as a likely default. The threshold analysis selected 0.25 based on validation-set F1 among the tested thresholds. This is a modelling choice, not a universal lending policy.

Table 2. Threshold comparison on the final test set

Threshold	Accuracy	Precision	Recall	F1
0.2500	0.7920	0.5265	0.5923	0.5574
0.5000	0.8213	0.6798	0.3632	0.4735

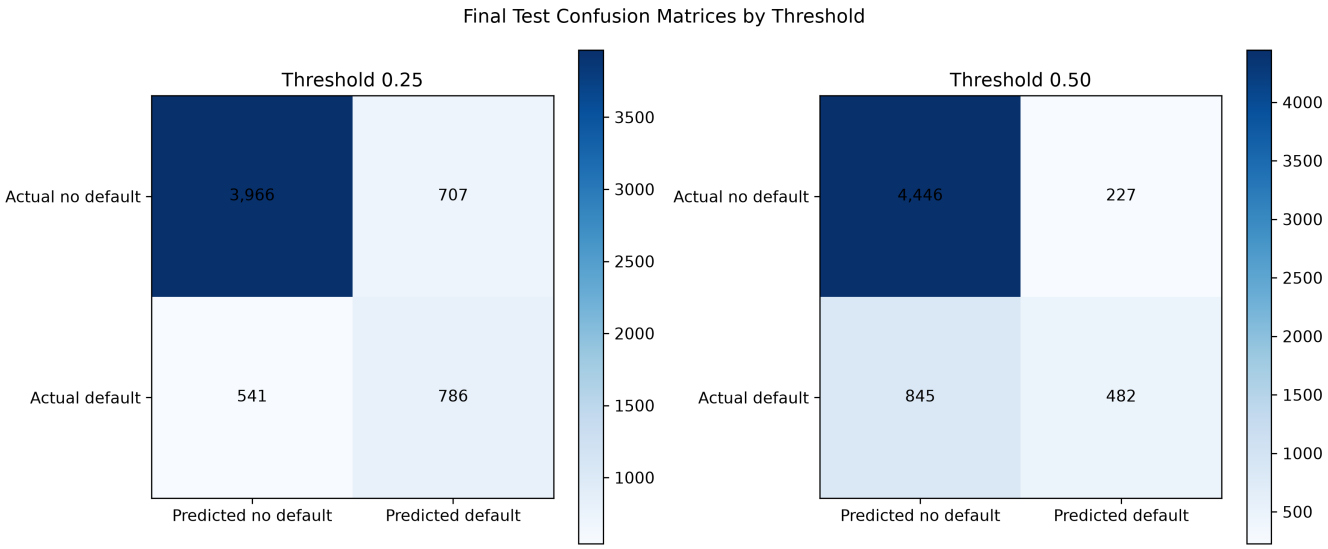


Figure 8. Performance comparison between the evaluated thresholds.

In the saved final test results, threshold 0.25 achieved recall of approximately 0.5923 and F1 of approximately 0.5574. Threshold 0.50 achieved recall of approximately 0.3632 and F1 of approximately 0.4735. The lower threshold identified more actual defaults but also generated more false positives.

The saved results indicate that threshold 0.25 produced 786 true positives and 707 false positives, compared with 482 true positives and 227 false positives at threshold 0.50. The appropriate trade-off depends on the costs and consequences of both types of error.

7. Subgroup Evaluation and Uncertainty

The project compares recall and false-positive rates across sex-coded categories and age groups. Wilson 95% confidence intervals show sampling uncertainty for subgroup recall and false-positive-rate estimates.

Table 3. Subgroup recall and false-positive-rate confidence intervals

Group type	Group	N	Recall	Recall CI low	Recall CI high	FPR	FPR CI low	FPR CI high
SEX	1	2358	0.6243	0.5832	0.6638	0.1804	0.1634	0.1988
SEX	2	3642	0.5696	0.5345	0.6040	0.1329	0.1210	0.1459
AGE_GROUP	29 and under	1907	0.5802	0.5344	0.6247	0.1494	0.1320	0.1687
AGE_GROUP	30-39	2274	0.6157	0.5697	0.6598	0.1432	0.1279	0.1601
AGE_GROUP	40-49	1274	0.5642	0.5072	0.6195	0.1585	0.1369	0.1827
AGE_GROUP	50-59	485	0.6053	0.5135	0.6902	0.1860	0.1497	0.2287
AGE_GROUP	60+	60	0.7059	0.4687	0.8672	0.0930	0.0368	0.2160



Figure 9. Recall estimates and confidence intervals by sex-coded group.

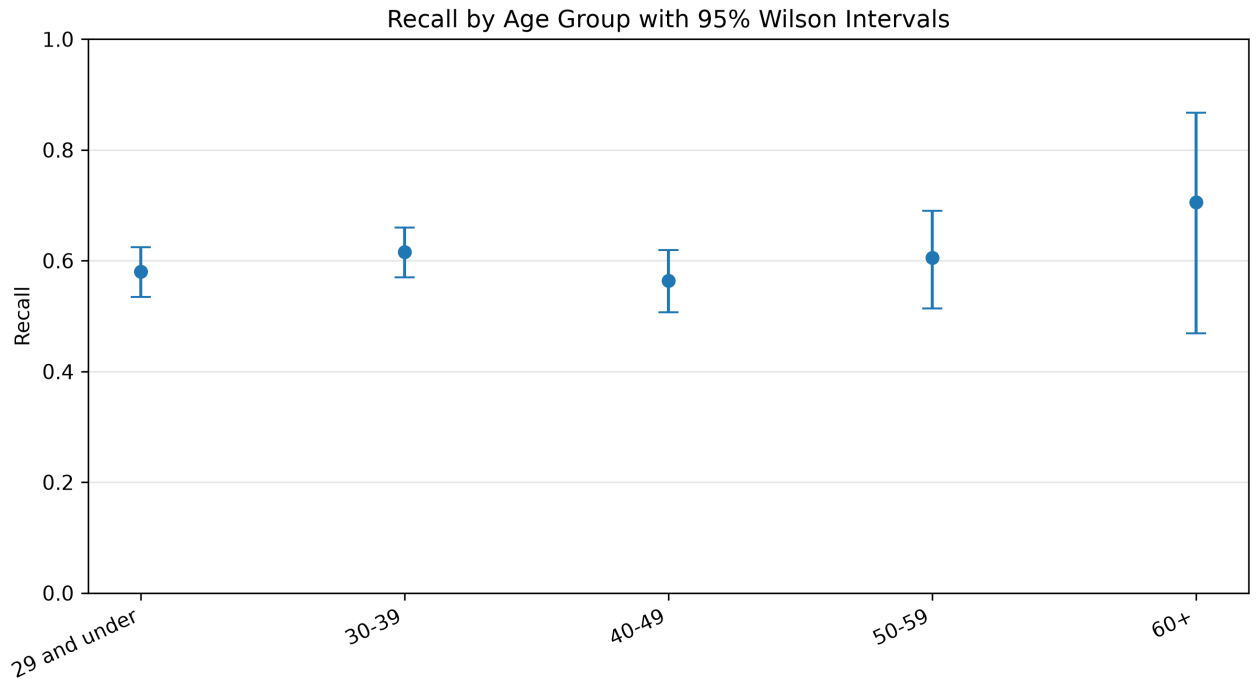


Figure 10. Recall estimates and confidence intervals by age group.



Figure 11. False-positive-rate estimates and confidence intervals by sex-coded group.

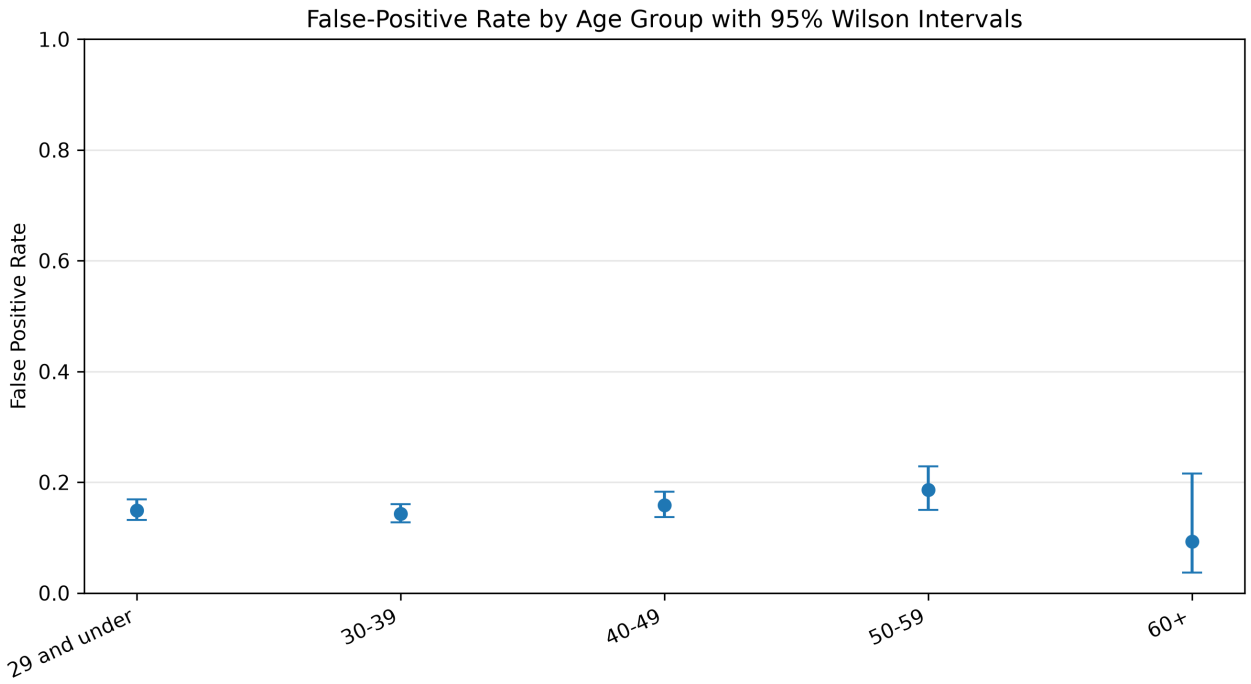


Figure 12. False-positive-rate estimates and confidence intervals by age group.

Subgroup differences warrant investigation but do not by themselves prove discrimination or establish fairness. The 60-and-over test subgroup contains only 60 records, producing particularly wide confidence intervals. The dataset's coded categories should be verified against the original documentation before substantive interpretation.

8. Limitations and Responsible Use

1. Population and time: The dataset is historical and specific to Taiwan. Results should not be assumed to generalize to other countries, institutions or time periods.

2. Evaluation independence: Earlier exploratory work used the same dataset. The final split does not replace independent external validation.

3. Threshold choice: A suitable threshold depends on the relative costs of missed defaults and false alarms, operational requirements and applicable rules.

4. Fairness: Confidence intervals describe uncertainty in subgroup rates. They are not a complete fairness assessment.

5. Explainability: SHAP describes model behaviour and should not be interpreted as causal evidence.

6. Research-only status: This project is educational research. It is not a validated or approved system for making real credit decisions.

9. Conclusion

The project demonstrates an end-to-end credit-risk research workflow covering data validation, exploratory analysis, model comparison, probability calibration, feature explanation, threshold analysis and subgroup uncertainty assessment. Gradient boosting achieved the strongest ROC-AUC in the saved final comparison, while threshold and subgroup analyses show why a single performance metric is not sufficient.

Future work should prioritize independent external validation, documented threshold-cost assumptions, additional fairness tests and careful review of data provenance before considering any real-world use.

10. References

1. Yeh, I. C., & Lien, C. H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications, 36*(2), 2473–2480. https://doi.org/10.1016/j.eswa.2007.12.020

2. University of California, Irvine. Default of Credit Card Clients dataset. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/350/default

3. Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems, 30.*

4. Scikit-learn developers. Scikit-learn documentation: model evaluation and probability calibration. https://scikit-learn.org/stable/

The references provide background for the dataset, model explanation method and evaluation concepts. Project metrics are taken from the locally saved experiment results.